

Day – 7 Software and Hardware tools.

Task – 1 Installing the Wireshark.

- 1. Transport Layer Security (TLS / TLSv1.3 or TLSv1.2)
- Why it dominates: Modern media apps encrypt almost all communications for user privacy and security. TLS handles the secure handshakes and encrypted data streams carrying your video chunks or photo feeds.
- 2. Transmission Control Protocol (TCP)
- Why it dominates: TCP manages the reliable, ordered delivery of data segments, ensuring that video buffers or image loads don't drop crucial packets as they travel across the network.

Task – 2 Use ping command.

- ? Open your Command Prompt (Windows) or Terminal (Mac/Linux).
- ? Run a ping test for Spotify:
- Bash
- ping www.spotify.com
- ? Run a ping test for Zomato:
- Bash
- ping www.zomato.com
- ? Look at the Average round-trip time in milliseconds (ms) at the bottom of each test.

Task – 3 List all process related to chrome.

- 1. Google Chrome (chrome.exe)
- Process Name: chrome.exe (Chrome runs multiple helper processes for tabs, extensions, and GPU acceleration).
- Memory Usage: Typically ranges from 100 MB to over 1 GB+, depending heavily on how many tabs you have open and what extensions are running. Browsers are notoriously memory-intensive because they isolate each tab into its own process for stability and security.
- 2. WhatsApp Desktop (WhatsApp.exe)

- Process Name: WhatsApp.exe (Built on Electron, which bundles a Chromium browser engine).
- Memory Usage: Usually ranges between 200 MB to 500 MB, even when running idle in the background, because desktop Electron apps keep web-based services active in memory for instant notifications.

Task – 4 Based on scenario.

- 1. Gateway / Router
- Status: Online
- IP Address: 192.168.1.1
- Device Name / Hostname: Gateway / Router (Manufacturer: TP-Link / Netgear)
- MAC Address: 00:1A:2B:3C:4D:5E
- 2. Your Local PC
- Status: Online
- IP Address: 192.168.1.15
- Device Name / Hostname: DESKTOP-WIN11 (Manufacturer: Intel Corporate)
- MAC Address: 54:BF:64:7A:8B:9C
- 3. Mobile Device
- Status: Online
- IP Address: 192.168.1.22
- Device Name / Hostname: Android-Phone (Manufacturer: Samsung Electronics)
- MAC Address: 88:36:6C:11:22:33
- How to Run it Yourself:

Task – 5 Based on scenario.

- Cable Tester: Use it to verify the continuity, correct pinout wiring, and overall integrity of an Ethernet cable after installation or repair.
-  Crimping Tool: Use it to firmly attach modular connectors (such as RJ45 plugs) onto the ends of twisted-pair or coaxial cables.
- Multimeter: Use it to test electrical parameters like voltage, resistance, and cable continuity to diagnose faulty power sources, adapters, or basic wire faults.
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